Teaching Statement

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My teaching philosophy is centered on creating a positive and engaging learning environment that supports the growth and development of all students. I believe that teaching is not just about imparting knowledge to students but also about inspiring them to inquire and learn independently. My goal as a teacher is to be a supportive and inspiring presence in my students' learning journey and to help them discover their own passions and abilities.

1 Teaching Experience

As a teaching assistant at the University of Illinois, I have gained valuable experience in teaching a variety of computer science courses. I have had the opportunity to work with students in foundational courses like 101 Computing and Data Structures, as well as more advanced classes like Applied Machine Learning. Through these experiences, I have learned the importance of adapting my teaching style to meet the needs and abilities of my students.

In my role as a TA, I have seen firsthand the joy and happiness that students experience when they learn and execute their first computer programming codes in foundational courses, and the sense of accomplishment they feel when they succeed in more advanced courses. To make the Applied Machine Learning course more interactive and engaging for students, we created Kaggle competitions as part of the assignments. This hands-on approach to learning allowed students to apply their knowledge in a real-world context and gave them the opportunity to experience the excitement and challenges of machine learning firsthand. These experiences have been incredibly rewarding for me, as I have seen the positive impact that hands-on learning and interactive assignments can have on student engagement and understanding.

I am committed to creating positive and engaging learning environments that inspire students to inquire, learn, and grow. I believe that by fostering a positive and supportive atmosphere in the classroom, I can help students feel comfortable taking risks and asking questions, and encourage them to pursue their own passions and interests. This approach has been recognized by my students, as I received an excellent teacher rating when assisting in the CS 101 - Intro to Computer Science course. In the future, I hope to continue building upon my experience as a TA and to continue creating positive and engaging learning environments for my students.

2 Teaching Plans

My goal is to inspire students from diverse backgrounds to discover the exciting and rewarding field of computer science. With my own personal experience transitioning from a background in civil engineering to computer science, I understand the importance of building a strong foundation in foundational subjects like probability and statistics, which are essential for success in machine learning and other advanced computer science courses. That's why I am particularly interested in teaching introductory courses in machine learning, computer vision, computational photography and probability and statistics to students with and without a prior CS background. I believe that these courses have the potential to open up new opportunities and inspire students to pursue careers in this dynamic field.

To make my courses engaging and relevant to my students' interests and career goals, I plan to incorporate interactive teaching methods such as group work, discussion, or problem-based learning. I also hope to integrate experiential learning activities or real-world examples into my

courses, as these can help students see the relevance of the material to their own lives and careers. Additionally, I plan to connect my courses to my own research interests and expertise in computer vision and learning-based computational photography, as this can bring a unique perspective and enthusiasm to the classroom and help students see the relevance of the material to real-world problems. Finally, I am excited to collaborate with other professors or professionals in the field to bring a diverse range of perspectives and experiences into the classroom. In summary, I am committed to creating a classroom culture that is positive, inclusive, and supportive of student learning and growth that helps them reach their full potential.

3 Mentoring Experience

In addition to my experience as a teaching assistant, I have also had the opportunity to mentor undergraduate, masters, and junior PhD students in their research projects. Through this experience, I have learned the value of guiding and supporting students as they pursue their own passions and interests. I have seen firsthand the creativity and potential of students given a bit of guidance, and have been humbled by their hard work and dedication.

As a mentor, I have had the privilege of working with a diverse group of students, and have seen the positive impact that mentorship can have on their development and growth. This experience has resulted in a number of notable achievements, including a publication at the Conference on Computer Vision and Pattern Recognition (CVPR) and a submission to the same conference, as well as the successful completion of a bachelor's and a master's thesis. I am proud of the accomplishments of these students and am grateful for the opportunity to have played a small role in their success.

In the future, I hope to continue mentoring and guiding aspiring students throughout my career. I believe that mentorship is a critical component of education, and that by supporting and guiding students, we can help them achieve their full potential and make a positive impact in the world.

4 Community Building Experience

As an educator, I am committed to fostering a sense of community and support among my students. I understand the importance of participating in a student reading group or community, as I have personally experienced the benefits of such groups during my own research journey. For the past three years, I have been organizing and leading a similar group for computer vision students at the University of Illinois. In addition to this, I initiated a weekly speaker series seminar that invites professors from academia and research scientists from industry to present their research. This series has now been formally recognized as an official one-credit seminar course at the University of Illinois.

I am also excited to be leading the organization of a workshop at the CVPR 2023 conference on the topic of "Scholars vs Big Models: How Should Academics Adapt?" This workshop will provide a valuable opportunity for researchers to share their knowledge and experiences on this important topic, and to discuss the challenges and opportunities of working with large models. Additionally, I am organizing a mini computer vision conference at the University of Illinois in Spring 2023, which will give students the chance to practice their research pitch and receive feedback on their work. These initiatives are aimed at building a strong and supportive community within the field of computer vision, and at providing students with the resources and support they need to succeed in their studies and research. I am confident that by creating these opportunities for collaboration and support, we can help students achieve their full potential and succeed in their academic endeavors.

5 Summary

I am thrilled to continue my teaching career and build upon my experiences as a teaching assistant, mentor, and community builder. As a teacher, I am dedicated to fostering a positive and supportive learning environment that empowers my students to reach their full potential. I believe in the importance of personalized and engaging learning experiences, and I strive to tailor my approach to meet the unique needs of each student. I am passionate about supporting and inspiring my students as they pursue their own learning and development, and I am committed to helping them discover their passions and abilities. I am confident that by creating a positive and engaging learning experience, I can make a meaningful impact on my students and help them succeed in their studies and beyond.